

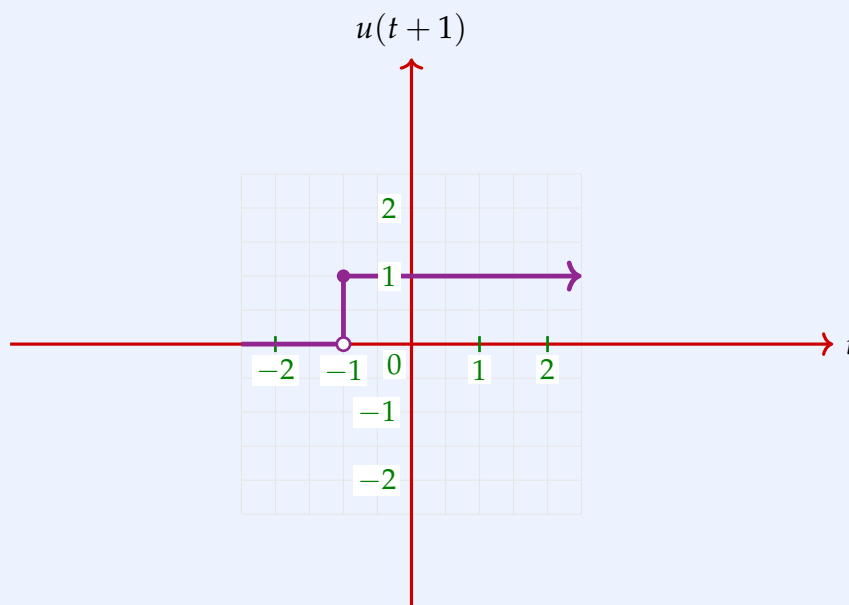
CPE 381 CLASSWORK 07: BUILDING BLOCKS FOR CONTINUOUS-TIME SIGNALS

Student Name (as in Canvas): _____

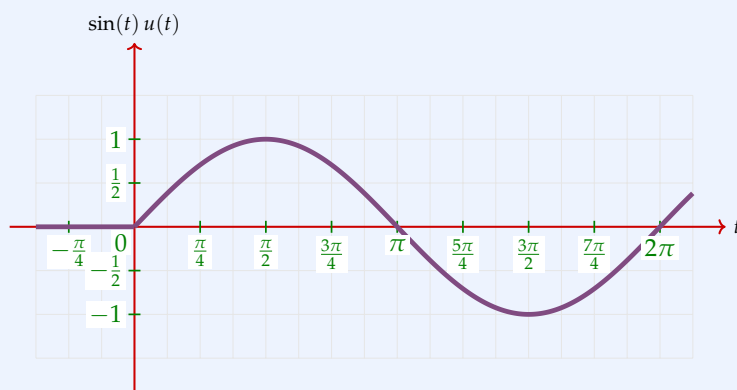
A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

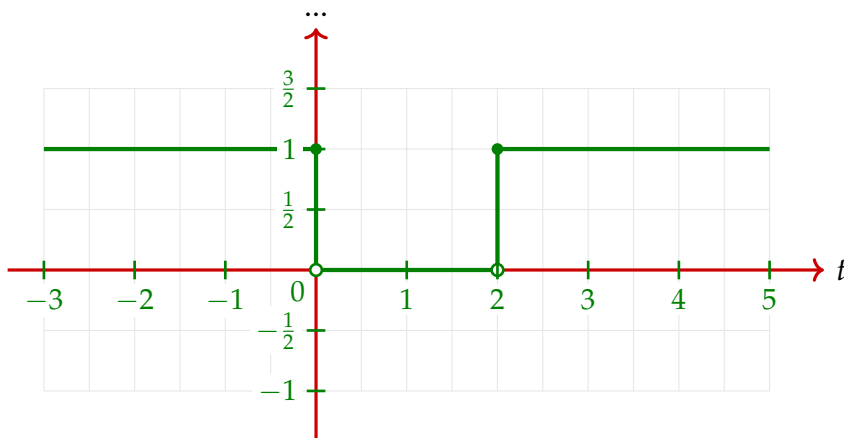
1. Assuming $u(t)$ is the unit-step function as discussed in the class, sketch $u(t+1)$.

Answer

2. Sketch $x(t) = \sin(t)u(t)$.

Answer

3. Write mathematical equation corresponding to the following graph.



Answer

$$u(t - 2) + u(-t)$$

4. Calculate the energy of the signal

$$z(t) = \begin{cases} 1, & 0 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

Answer

Energy of the signal is given by

$$E_z = \int_{-\infty}^{\infty} |z(t)|^2 dt = \int_0^{10} |1|^2 dt = \int_0^{10} 1 dt = [t]_{t=0}^{t=10} = 10 - 0 = 10$$

Note that integral bounds changes from $\pm\infty$ to $[0, 10]$ due to signal's definition provided in the question. The signal is zero outside $t = [0, 10]$.